Lab 5:

Implement WordCount program on Hadoop framework

```
// WCDriver.java
import java.io.IOException;
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.*;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {
    public int run(String[] args) throws IOException {
        if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
        }
        JobConf conf = new JobConf(WCDriver.class);
        conf.setJobName("WordCount");
        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));
        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);
        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);
        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);
        JobClient.runJob(conf);
        return 0;
    }
    public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
    }
}

// WCMapper.java
public class WCMapper extends MapReduceBase
        implements Mapper<LongWritable, Text, Text, IntWritable> {
    public void map(LongWritable key, Text value,
            OutputCollector<Text, IntWritable> output, Reporter reporter)
            throws IOException {
        String line = value.toString();
        for (String word : line.split("\s+")) {
            if (word.length() > 0)
                output.collect(new Text(word), new IntWritable(1));
        }
    }
}

// WCReducer.java
public class WCReducer extends MapReduceBase
```

```
        implements Reducer<Text, IntWritable, Text, IntWritable> {
    public void reduce(Text key, Iterator<IntWritable> values,
            OutputCollector<Text, IntWritable> output, Reporter reporter)
            throws IOException {
        int count = 0;
        while (values.hasNext())
            count += values.next().get();
        output.collect(key, new IntWritable(count));
    }
}

// Execution
$ start-all.sh
$ hdfs dfs -mkdir /Lab06
$ hadoop fs -copyFromLocal -f /home/hadoop/Desktop/file1.txt /rgs/test.txt
$ hadoop jar /home/hadoop/Desktop/Word_Count.jar wordcount.WordCount \
    /rgs/test.txt /output
$ hadoop fs -ls /output
$ hadoop fs -cat /output/part-00000
```

```
// hadoop fs -ls /output:
Found 2 items
-rw-r--r--  1 hadoop supergroup   0  2026-05-03 10:15  /output/_SUCCESS
-rw-r--r--  1 hadoop supergroup  69  2026-05-03 10:15  /output/part-00000

// hadoop fs -cat /output/part-00000:
are      1
brother  1
family   1
hi       1
how      5
is       4
job      1
ls       4
sister   1
you      1
your     4
```